



# Investigation report

## Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads - Probable cause report

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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## Executive summary

### What happened

On 07/07/2026 at 18:12:04 local time, MV Tarrakin Star grounded on Boolarong Bank at 6.8 kt. The vessel is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. It was inbound on service TF18 through the Weerona Channel at Kanooka Heads, in darkness on a falling tide, carrying 38 passengers, a master, a deckhand and a kiosk attendant. It stopped within 11 m, heeled 4° to starboard, 74 m west of the charted centreline and 54 m outside the dredged channel.

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. Damage was substantial:

- Starboard hull bottom plating set up over 4.1 m, with a 180 mm split in way of the No. 2 void, which flooded to approximately 260 mm.
- Starboard propeller blades and shaft bent.
- Starboard rudder stock distorted and skeg torn away.

All passengers were ashore by 19:26. The vessel refloated at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

### What the investigation found

The probable cause was that the chart plotter display very likely froze during a night transit for which the operator had no night passage plan and no hold point. The vessel continued at about 8 kt while the No. 4 beacon, the only remaining means of fixing, was very likely screened from the channel by mangrove canopy. It left the dredged channel and tracked onto the bank.

The plotter froze at 18:07:06. A stop in 92 m was available throughout the 114 seconds before the vessel crossed the channel's western edge, but nothing prompted the master to consider one. Continuing at speed very likely contributed. The shallow alarm, set at 0.9 m under the keel since 2019, very likely contributed: working as set, it gave 16 seconds' warning, covering 58 m against a crash-stop distance of 92 m.

- **Timetable change.** The winter timetable moved TF18 into darkness without reopening the route risk assessment, which restricted the channel to daylight transits. This likely contributed.
- **Plotter defect reporting.** The plotter had frozen twice that morning. The fault was passed on verbally at the change of master and not written down. Handover practice did not require a written brief, and the manufacturer's firmware fix issued in 11/2024 had never been applied. The omission and the practice each likely contributed. However, a single entry that afternoon would not by itself have stopped TF18; on that reading the handover rule is a fleet-wide deficiency rather than a cause of this grounding.
- **Beacon sightline.** Clearing was deferred as unfunded at three works rounds, and the canopy grew above the lantern. This very likely contributed.
- **Beacon range.** Solar conversion halved the beacon's nominal range and no operator was told; the passage plan still showed 6 M. This likely contributed.
- **Regulatory scope.** The applicable code placed no availability, recording or notification duty on this channel's aids. This likely contributed, and is the least firmly held contributing factor.

Two matters did not influence the occurrence but are reported as risks: the forward saloon lifejacket locker cannot be reached from 22 of its 34 seats without passing behind the kiosk counter, and the

master was on day nine of twelve, on a split duty. A low water level and a steering or propulsion failure were ruled out. Whether new foreshore floodlighting behind the beacon affected its detection could not be established.

After contact, the master stopped the engines and did not attempt to refloat the holed vessel on a falling tide. This very likely reduced the risk to passengers.

## What can be done

The investigation identified seven safety issues: the timetable change not reopening a live daylight-only restriction; no night passage plan or hold point for loss of the primary means of fixing; verbal, unchecked defect handover; a code imposing no duties on aids outside declared shipping channels; a beacon range change never passed to users; sightline clearance deferred for three years; and a modification route that relocated a lifejacket locker without re-survey, with the same counter fitted to two other vessels.

Lessons:

- **Operators:** check whether a change invalidates an existing risk assessment, not only whether it introduces a new hazard. Give crews a written point at which to stop when the primary means of fixing is lost. Ensure intermittent faults reach someone ashore.
- **Authorities:** an aid that is lit but cannot be seen from the channel is not an effective aid.

No safety action had been recorded.

## Factual information

### History of the occurrence

Times are local, 10 hours ahead of UTC.

MV Tarrakin Star changed masters at the town wharf at about 13:05 on 07/07/2026. The outgoing master told the relieving master the chart plotter had frozen twice that morning, recovering by itself each time. The relieving master replied that it had done the same to him the month before. No defect book entry was made for 07/07/2026 before the occurrence.

Service TF18 was the last of the day and the thirty-first evening return of the winter timetable, all in darkness since 01/06/2026. It departed Muntari Island at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant. Sunset was at 17:02; civil twilight ended at 17:29.

The vessel entered the Weerona Channel from the north abeam the No. 1 beacon at 18:03:58. At 18:04:14 the master sent the deckhand to collect fares; the deckhand stayed in the saloons for the rest of the transit. At 18:05:30 the vessel was on the charted centreline abeam the No. 2 beacon at 8.4 kt, course 195°.

At 18:07:06 the plotter display froze and stopped accepting touch input. At 18:07:44 the master said "frozen again". He continued by eye, looking for the No. 4 starboard-hand beacon on the channel's western side, saying at 18:08:31 "Right. Where's four." At 18:09:40 the audio recorded a sound consistent with a window or door sliding open.

From 18:07:06 course over ground increased from 195° to 200° and the vessel set west at an average of 12 m per minute, crossing the western edge of the dredged channel at about 18:09:00. At 18:10:02, 39 m west of the centreline, the master said "There's four. Keep her over." The vessel was over Boolarong Bank at about 18:11:00.

The shallow alarm sounded at 18:11:48; the master moved both throttles astern at 18:11:52. At 18:12:04 the vessel contacted Boolarong Bank at 6.8 kt and stopped within 11 m, heeled 4° to starboard, 74 m west of the centreline and 54 m outside the dredged channel, on a falling tide. About 9 seconds of grinding and vibration were recorded. The master stopped both engines at 18:12:31.

At 18:12:40 the master told passengers by public address that the vessel had touched the bottom. At 18:13:04 he broadcast a pan pan on VHF channel 16 to Kanooka Heads Port Control, giving the position, 38 passengers and three crew, no injuries reported, no pollution, and that he was not attempting to refloat. Port Control acknowledged at 18:13:29 and called out the port workboat.

The deckhand sounded the No. 2 starboard void at 18:16 and found water; the master reported approximately 75 mm at 18:17:05 and sounded every ten minutes thereafter. All passengers were in lifejackets by about 18:22; issue in the forward saloon was delayed because its locker is behind the kiosk counter. The workboat came alongside at 18:41, and all 38 passengers were reported ashore at 19:26.

The vessel refloated on the rising tide at 00:20 on 08/07/2026 with approximately 260 mm in the No. 2 void, was towed to the town wharf and withdrawn from service. The channel was closed to commercial traffic for 19 hours.

### Injuries

No serious injuries. One passenger reported a grazed shin; two were treated at the wharf for distress.

## Personnel information

**Master.** Certificate of competency, master less than 35 m, since 2012. Master for 14 years, 11 on this route with Tarrakin Ferries; about 19 years in commercial marine operations; approximately 4,100 channel transits. Route check satisfactory 14/02/2026. Day nine of a twelve-day tour, on a split duty of 05:40 to 08:40 and 13:10 to 19:20; slept 6 h 15 the night before. He described himself as tired, normal for a split, and said he was not fighting to stay awake and not slow.

**Outgoing master.** Master for 9 years, 4 on this route; about 15 years in commercial marine operations; approximately 1,400 channel transits and 5 of the 31 darkness transits. Route check satisfactory 09/11/2025. Worked 05:40 to 13:05, day six of his tour.

**Deckhand.** Seven months with the operator, in a first winter season on the route. Held a certificate of safety training and a coxswain grade 2; about 3 years in commercial marine operations. Familiarisation and drills completed 12/2025. Had not been shown the channel marks and did not know which beacon was the No. 4.

## Vessel and equipment information

MV Tarrakin Star is a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers. Annual survey completed 11/03/2026 with no outstanding conditions; last slipped 03/2025.

The defect book held eleven entries in the preceding 90 days, none concerning the plotter, GPS antenna, echo sounder, steering, engines or propulsion controls. The maintenance system held no work order, fault report or correspondence on the plotter in 2026.

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024. Firmware issued 11/2024 addresses cause code 0x51 and had not been applied. The operator did not hold the bulletin and had no subscription to the manufacturer's bulletins.

The shallow alarm was set to 0.9 m under the keel, unchanged since commissioning in 2019. The trials crash-stop distance from 8 kt is 92 m.

## Damage

- **Starboard hull.** Bottom plating set up over 4.1 m between frames 6 and 11; 180 mm split in way of the No. 2 void.
- **Starboard propeller and shaft.** All three blades bent aft; shaft bent 1.4 mm.
- **Starboard rudder and skeg.** Rudder stock distorted 3°; skeg torn away.
- **Port side.** No damage to hull, propulsion or steering.

## Meteorological information

Both masters' accounts and the master's report describe a clear evening, light westerly and no sea in the entrance. The transit was in full darkness.

## Recorded data

Sources were the shore AIS receiver log, the engine monitoring recorder and the steering data.

| Time     | West of centreline | North of No. 4 | Speed over ground | Course over ground |
|----------|--------------------|----------------|-------------------|--------------------|
| 18:05:30 | 1 m                | 0.90 nm        | 8.4 kt            | 195°               |
| 18:07:06 | 3 m                | 0.68 nm        | 8.4 kt            | 196°               |
| 18:08:31 | 12 m               | 0.47 nm        | 8.0 kt            | 199°               |
| 18:09:00 | 20 m               | 0.40 nm        | 8.0 kt            | 199°               |

|          |      |         |        |      |
|----------|------|---------|--------|------|
| 18:10:02 | 39 m | 0.25 nm | 7.9 kt | 200° |
| 18:11:48 | 70 m | 0.03 nm | 7.1 kt | 198° |
| 18:12:04 | 74 m | abeam   | 6.8 kt | 197° |

Rate of turn peaked at 2.1° per minute at 18:07:40 and returned to zero by 18:08:50. Speed fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement.

Engines ran at 1,780 and 1,784 rpm until 18:11:52 with no alarm, temperature or pressure excursion, or throttle movement. Steering hydraulic pressure stayed at 96 to 99 bar (normal band 90 to 105 bar); rudder angle followed helm demand within 1.5°. At 18:11:52 both throttles moved astern together; astern thrust had not reached full at contact.

The plotter event log recorded:

- 10:42 and 11:17: watchdog resets, cause code 0x51, display process not responding.
- 18:07:06: display process not responding, no reset recorded.
- 18:16: power removed.

The shallow alarm sounded once, 16 seconds before contact, in which the vessel covered 58 m.

## Site information

The position, surveyed on 08/07/2026, lies 74 m west of the charted centreline, 54 m west of the channel's western edge and 34 m west of the No. 4 beacon. The bed at contact was 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70. Furrows consistent with the starboard hull and skeg ran 11 m from first contact on 197°. The skeg was recovered from the bank.

## Organisational and management information

**Timetable change.** The winter timetable took effect on 01/06/2026. Both masters and the deckhand said only a new roster or new times came with it. The operator had no night passage plan for the channel. The passage plan names a speed, a course and three beacons; neither it nor OPS-3 addresses who is on the bridge. The deckhand said no bridge lookout had ever been kept at night on this run.

**Defect reporting.** Both masters said they did not regard a fault that had cleared as something to enter. The outgoing master believed the plotter had frozen three or four times in the past year, none written up. The operator's "navigation hazard" reporting category had not been used on this route in three years, and no master interviewed knew it existed.

**No. 4 beacon.** Light character Fl G 2.5s. In 11/2025, under the Council's waterways energy programme, it was converted from mains to a solar lantern, reducing nominal range from 6 M to 3 M. The hydrographic service was notified on 04/02/2026; the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. The operator's passage plan appendix still showed 6 M.

WO-118, raised 09/2023, recorded mangrove regrowth on Weerona Islet screening the beacon from the northern approach; it was deferred as unfunded at the 2023, 2024 and 2025 works rounds. WO-176, raised 03/2026, recorded the sightline fully closed. The Port Authority's file note placed the islet outside the 50 m scope of schedule 2.4 of the services agreement, so the work fell to the general waterways vote.

The beacon was logged "observed, operating" monthly from 01/2024 to 07/2026 from the town wharf, 2.1 nm south, where the islet is not in the line of sight. No defect or notice had ever been recorded against it. The Authority had never provided the aids to navigation register to Tarrakin Ferries and had no record of any operator enquiry about the channel's aids.

Both masters said the beacon had become harder to see over two years; neither had reported it. The master said, "There's no form for the beacon looks dim." Both referred to new car park floodlighting behind the beacon. The master said twice he could not say whether the lighting was why he could not see the beacon, only that it was harder to see than it used to be.

## Tests and research

**Chart plotter.** An approved contractor bench tested the plotter on 21/07/2026 for 46 hours with the same antenna and cabling. The fault did not recur; no defect was found.

**Steering gear.** Rams, tiller arms, hydraulics and autopilot were removed and tested on 12/07/2026. All were serviceable and passed a lock-to-lock timing test.

**Range and visibility.** On 18/07/2026 a check from the workboat on the inbound track at the same time of night measured luminous range at 2.6 nm. The light was not visible between 0.84 nm and 0.34 nm north of the beacon, the sightline passing through the Weerona Islet canopy 240 m north-west of it. The canopy measured 5.8 m, against 2.1 m in the 2019 aerial survey. The lantern is 4.2 m above chart datum.

## Additional information

**Forward saloon lifejacket stowage.** The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon locker holds 40 adult jackets behind a kiosk servery counter fitted in a 2019 refit under a minor modification notice, which did not trigger re-survey. The 2011 approved arrangement had an open bench with the locker face clear.

Measurements on 16/07/2026 found 22 of the 34 forward saloon seats cannot reach the locker without passing through a 0.6 m gap behind the counter. The muster placard directs passengers to a locker not visible from the seats. The same counter is fitted to the operator's two other 24 m vessels. The deckhand has submitted a safety report on the arrangement.

## Analysis

### External Influences

#### **Aids to navigation on this waterway carry no availability standard and no duty to tell users anything**

The Waterways Aids to Navigation Code WAN-2 sets two tiers of obligation. Under section 4.2, an authority responsible for an aid on a declared shipping channel:

- must record the aid's availability,
- must achieve "an availability of not less than 99.0 per cent measured annually", and
- must "notify a change to the aid's character, position or range by notice to mariners before the change takes effect".

The Weerona Channel is not a declared shipping channel. Section 4.3 therefore applied to all six of its beacons. That section requires only that the authority "be able to demonstrate that the aid is inspected at intervals it has determined and recorded."

The services agreement between Kanooka Shire Council and the Kanooka Heads Port Authority set those intervals:

- a daylight inspection each quarter, and
- a monthly observation from the town wharf.

The agreement limited vegetation work to within 50 m of an aid. It stated that there "is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." The Authority was not in breach of the code or the agreement. That these arrangements existed and applied to the No. 4 beacon was virtually certain.

Each of the mechanisms the code provides for carrying an aid's condition to its users stopped at the boundary of this channel. None of them applied to the No. 4 beacon:

- a measured and recorded performance,
- an availability target, and
- advance notice of a change in range.

The lantern was converted to solar in November 2025. The conversion reduced its nominal range from 6 to 3 nautical miles. The Authority lodged a notification with the hydrographic service on 04/02/2026, and the chart correction was still pending on the night. No local notice to mariners was issued and no operator was told. The operator's passage plan appendix still showed the No. 4 beacon at 6 M. The aids to navigation register had never been provided to the operator. The only remaining channel for information about the beacon was informal exchange between masters.

The agreement's own terms narrowed what the Authority's checks could detect:

- The monthly observation was made from the town wharf, 2.1 nm south of the beacon. Weerona Islet is not in the line of sight from there. The beacon was logged as observed and operating every month from January 2024 to July 2026.
- The islet lies 240 m from the beacon, outside the 50 m scope of the agreement. The Authority's file note recorded that clearing work on the islet therefore competed with boat ramps and pontoons in the general waterways vote.

Evidence against a causal link was also examined, and it was substantial.

First, nothing in the code prevented the Authority from doing any of the following:

- notifying operators,

- trimming the mangroves, or
- measuring the light.

The Authority's choices not to act were made on grounds of funding and priority, not because the code required them. On that view, the code's scope was background rather than cause.

Second, a 99.0 per cent availability target would not have detected the hazard at all. The lantern was lit and working throughout. The problem was that it could not be seen from the channel between 0.84 nm and 0.34 nm north of the beacon. No availability figure measures that.

The investigation accepted the second point. The availability standard was not relied on in reaching the conclusion.

The first point was weighed against the notification duty in section 4.2. That duty was the one mechanism identified that would have placed the range reduction before the operator, whose passage plan still carried the superseded figure on the night. Had the channel been subject to section 4.2, a notice to mariners would have been required before the conversion took effect. Discretion would not have been left to the Authority's priorities. On the balance of that evidence, the investigation found the influence likely.

This finding is held with less confidence than the investigation's other conclusions on causal influence, and it should be read that way. The opposing argument was not disproved. It was outweighed.

The code's scope was found to have led to two organisational conditions, both argued under Organisational Influences:

- the solar conversion that halved the beacon's nominal range without any operator being told, and
- the deferral of the sightline clearing, unfunded, across three works rounds.

The absence of an availability standard or notification duty for aids on the Weerona Channel was likely to have contributed to the grounding. It allowed a reduction in the No. 4 beacon's range to go unnotified to the channel's users, and it left the checks on the beacon unable to detect that it was screened from the northern approach.

## Organisational Influences

### **Winter timetable moved the last service into darkness without reopening the route risk assessment**

RRA-07 issue 2 restricted the Weerona Channel to daylight. Its wording was explicit: "A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

CHG-2026-014 moved TF18 from 16:15 to 17:35 from 01/06/2026. From that date, TF18 transited the channel after the end of civil twilight on every day of the winter timetable. The change record assessed itself as "timetable change only", recorded "No new hazard identified" and answered "No" to whether a route risk assessment review was required. It did not refer to the restriction.

The Operations Manager approved the change. That approval sat one level below the approver the restriction names. RRA-07 was not reopened and no night passage plan was written. The General Manager Operations' approval was neither sought nor given. TF18 then ran 31 times in darkness before the occurrence, and the master made 26 of those transits.

The change process asked whether the route, vessel, crew qualifications and berths were new. None was. It did not ask whether the conditions the existing assessment relied on still held, and daylight was one of those conditions.

Both documents were current controlled copies. The Operations Manager's account of how he understood the restriction was given verbally, not in formal interview. It was extremely likely that the change was made without the review the restriction required.

A review against a night operation would have reached the reach between the No. 2 and No. 4 beacons, which RRA-07 already named as a hazard. It would have required a night passage plan before the first night service. Such a plan might have said no more than the daylight one, and the influence is not put higher for that reason. The Risk Controls subsection argues the absence of a night passage plan and of any hold point on loss of the plotter.

It was likely that approving the winter timetable without reopening RRA-07 led to the channel being transited at night with no night passage plan and no hold point.

### **Beacon converted to solar, nominal range halved, and no operator was told**

The No. 4 beacon was converted to a solar lantern in 11/2025. The new lantern was rated at 3 nautical miles, against the 6 of the lantern it replaced. The character, Fl G 2.5s, did not change, so nothing in the light itself showed the change. The hydrographic service was notified on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. The operator's current passage plan listed the beacon as "Fl G 2.5s, range 6 M". The luminous range measured on 18/07/2026 was 2.6 nm. These matters were documentary or independently measured, and virtually certain.

The master was therefore working to a plan that described a light with more than twice the range of the one fitted. A weaker light is harder to acquire. The master searched for the beacon for 91 seconds, from 18:08:31 to 18:10:02.

Against that, the vessel covered 0.47 nm to 0.25 nm north of the beacon during the search. For most of that stretch, down to 0.34 nm, the light was screened entirely, and a screened light is not made visible by being brighter. The reduced range could bear only on the final part of the search, after the light opened past the islet, and on what the master expected to see.

Whether shore lighting behind the beacon affected its detection could not be resolved and is discussed separately. Nothing from that matter is carried into this conclusion.

It was likely that the unannounced reduction in the beacon's range contributed to the No. 4 beacon not being available to the master as a visual reference during the transit.

### **Clearing of the beacon's sightline deferred unfunded for three works rounds**

WO-118 was raised in 09/2023. It identified mangrove regrowth on Weerona Islet screening the beacon from the northern approach, between about half a mile and a quarter mile. It recommended trimming to 2 m on a three-yearly cycle, at an estimated \$18,500. The work was deferred as unfunded at the 2023, 2024 and 2025 works rounds.

WO-176, raised in 03/2026, recorded the sightline as fully closed from the north. The islet lay 240 m from the beacon, outside the 50 m scope of schedule 2.4 of the services agreement. The work therefore competed with boat ramps and pontoons in the general waterways vote, rather than with aids to navigation. The deferral was documented in the Authority's register and the screening was measured on site, so it was virtually certain.

The canopy was 2.1 m in the 2019 survey. On 18/07/2026 it measured 5.8 m, against a lantern height of 4.2 m. Over the screened arc the light was neither dim nor defective; it was absent.

The screened arc ran from 0.84 nm to 0.34 nm north of the beacon. The key positions were:

- the plotter froze at 0.68 nm;
- the master began looking for the beacon at 0.47 nm;
- the vessel crossed the western edge of the channel at about 0.40 nm;
- the master called the beacon at 0.25 nm.

All of the search before the vessel left the channel fell within the screened arc.

The monthly check recorded the beacon as operating. It was made from the town wharf, 2.1 nm south, from which the islet is not in the line of sight, so it could not detect the screening.

The master might not have picked the light up in that minute and a half even had it been visible. This is the only allowance against influence.

It was very likely that the deferral of the sightline clearance contributed to the No. 4 beacon not being visible from the channel over the half mile before the vessel left it.

## Risk Controls

### **No night passage plan for the channel, and no hold point if the primary means of fixing is lost**

The transit was made under OPS-3 and its Appendix A, issue 2, the current issue. OPS-3 3.4 requires the transit to be "monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon." The appendix names the plotter as the "Primary means of position fixing" and closes: "**Night transits are not scheduled on this route.**" It contains no guidance on losing the plotter and names no abort or hold point. The operator confirmed that no night passage plan for the Weerona Channel existed. The absence of the plan and of any hold point is virtually certain.

The appendix did name a secondary means: the visual transit of the No. 4 beacon with the town wharf leading lights. That provision did not work as written. The leading lights lie 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third. When the plotter froze at 18:07:06, the vessel was 0.68 nm north of the No. 4 beacon, so the secondary means depended on the beacon alone. OPS-3 3.6 requires a reduction in speed and consideration of anchoring, but only "in reduced visibility". The visibility on the night was not reduced, so 3.6 did not apply.

The master had about 114 seconds between the freeze and the vessel crossing the channel's western edge. The water was flat and there was no other traffic. A named hold point would have been available for the whole of that period. Both masters interviewed told the investigation, independently, that the correct response was to stop, and that nothing in the operator's procedures said so. The evidence against was that a master who did not think of stopping unprompted might also not have applied a written rule under the same pressure. The investigation accepted that possibility and weighed it against the time and conditions available.

How the timetable change led to night transits without a plan, and the master's decision to continue at 8 knots, are discussed in their own subsections.

The absence of a night passage plan and of any hold point for loss of the plotter very likely contributed to the transit continuing after the plotter froze.

### **Change-of-master defect brief is verbal, with nothing checking that what is said gets written down**

OPS-9 9.1 requires immediate reporting only for a defect that makes the vessel unseaworthy or affects a statutory item. The chart plotter is not a statutory item on a class 1D vessel on this route, although the passage plan names it as the primary means of position fixing. Other defects fall under 9.2, which requires them to be "entered in the vessel defect book at the end of the shift in which they are found". Such entries are reviewed by the Operations Manager weekly. Under 9.3 the outgoing master briefs the relieving master on "any defect". Nothing requires that brief to be written down, and nothing requires the relieving master to confirm that a defect mentioned in it has been booked.

Both masters independently described the same practice: the book is for what is broken at handover, and what has recovered is passed on verbally. The plotter's log recorded freezes at 10:42 and 11:17 that morning, and both times the plotter recovered by itself. At the change of master at 13:05 the fault was passed on verbally. No entry of any kind was made for the day. The outgoing master believed the plotter had frozen three or four times on this vessel in the previous year, to him and to others, and that none of those freezes had been written up. The maintenance system held no plotter entry of any kind for 2026. The operator did not hold the manufacturer's firmware bulletin issued in 11/2024, which addressed cause code 0x51, and it had no subscription to those bulletins. That the handover practice existed is virtually certain. It explains why the fault was not entered. It does not excuse the omission.

The investigation considered two readings of the practice's influence.

- **Against:** an entry made at 13:05 would have been reviewed at the end of that week, not that afternoon, and TF18 would have sailed either way. On that reading, the handover practice is a genuine weakness across the fleet but not a cause of this grounding.
- **For:** the relevant matter is not the single missed entry on the day. Several freezes over a year never reached anyone ashore who could act on them. As a result, the firmware fix was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance.

The investigation accepted the second reading while recognising the force of the first.

The verbal change-of-master defect brief, with no check that defects were booked, likely contributed to the plotter fault going unrecorded and unrectified before the transit.

### **Forward saloon lifejacket locker not reachable from most of the saloon it serves**

The forward saloon locker holds 40 adult lifejackets. It is mounted in the bulkhead behind the kiosk servery counter. A survey on 16/07/2026 found that the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter through a 0.6 m gap. With three people standing at the counter, as there were on the night, that gap is the only route. The muster placard directs passengers to that locker, which cannot be seen from the seats. While issuing lifejackets, the deckhand had to reach the locker around the servery and reported to the master that "the locker's behind the counter and there's people standing at it." The condition is virtually certain.

It is unlikely that the locker position had any influence on the occurrence. The grounding happened independently of where the locker was. All 38 passengers were in lifejackets within about eight minutes of the first instruction and ashore within 74 minutes of contact.

The condition matters beyond this occurrence. The 2011 approved arrangement had an open bench in that position, which left the locker face clear. The counter was fitted in a 2019 refit under a minor modification notice. That notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels. In a fire, a night

evacuation into the water, or with a crew member injured, passengers could not equip themselves without crew help.

The inaccessible forward saloon lifejacket locker, created by a modification route that did not consider safety equipment, very likely increased risk across the class.

## Local Conditions

### **Chart plotter display froze during the channel transit and did not recover**

Three independent sources agree on the freeze. The plotter's event log records the display process not responding at 18:07:06 with no reset. The bridge audio follows within seconds: "Oh, come on" at 18:07:11, repeated screen taps at 18:07:26, and "(unintelligible) frozen again" at 18:07:44. The master's account describes a stationary vessel symbol and a screen that would not accept touch input.

The failure could not be reproduced. A 46-hour bench test with the same antenna and cabling found no defect. The evidence for the freeze therefore rests on the unit's own log rather than on a failure shown independently. Two points support that log. Cause code 0x51 is a fault known to the manufacturer. Firmware issued in 11/2024 addresses it, and that firmware had not been applied to this unit. How the morning freezes were handed over is analysed under Risk Controls.

The vessel held the centreline while the plotter worked. It began to leave the centreline from 18:07:06, and the master began searching for the beacon 85 seconds after the freeze. From that point the transit was being made without the primary means of position fixing named in the passage plan. The decision to continue at eight knots is analysed under Risk Controls.

The chart plotter display very likely froze during the transit and did not recover, and this extremely likely contributed to the vessel continuing without a means of fixing its position.

### **No. 4 beacon not visible from the channel over the half mile before the vessel left it**

The range and visibility check on 18/07/2026 was made from the inbound track at the same time of night. It found the No. 4 beacon was not visible from the channel between 0.84 nm and 0.34 nm north of it. Over that stretch the sightline passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy measured 5.8 m, against a lantern height of 4.2 m.

The vessel's own track places the master's search inside that arc. The vessel was 0.47 nm north of the beacon at "Right. Where's four" (18:08:31) and 0.25 nm at "There's four. Keep her over" (18:10:02). These remarks were 91 seconds apart, and the vessel crossed the channel's western edge between them. Both masters raised the beacon without being asked. The only qualification is timing: the canopy was measured eleven days after the grounding, not on the night.

With the plotter frozen, the beacon was the only means of fixing left. It was unavailable for the whole period in which the vessel was set west. The vessel had begun to set before the master started looking, which is why the influence is put no higher than very likely. The master was not failing to look; the light could not be seen. The port decisions that allowed the screening are analysed under Organisational Influences and External Influences.

The No. 4 beacon was extremely likely not visible from the channel over the stretch in which the master searched for it, and this very likely contributed to the vessel leaving the dredged channel undetected.

### **New foreshore floodlighting behind the beacon reduced the master's ability to pick it up**

This matter was examined and could not be resolved.

**What supports it.** The master described looking into a wall of light, with the beacon lost against the town and the new car park. The outgoing master, independently, described the light as washed out from the approach. The 18/07/2026 survey found 14 unshielded LED floodlights on 12 m columns at the new boat ramp. Each presents about 4,800 candela toward the channel. They were commissioned on 22/05/2026, ten days before the winter timetable began. From 0.5 nm north of the beacon, the array lies behind it within 4 degrees of bearing and 1.5 degrees of elevation. The array is continuously lit, while the beacon shows a 0.4-second green flash every 2.5 seconds. The Port Authority's aids to navigation officer was not consulted and no assessment was made.

**What counts against it.**

- The only position where the alignment was measured, 0.5 nm north, lies inside the screened arc. There the beacon would have been hidden even against a dark background.
- The master found the light once the vessel passed the southern limit of that arc, with the floodlights unchanged.
- No measurement exists from the night itself.
- No detection threshold has been established for this flash against this background.

**Why it could not be settled.** The masters' accounts and the measured alignment are consistent. However, the evidence cannot separate the effect of the floodlights from the effect of the canopy. The question may not be answerable while the mangroves remain. The alignment is nonetheless real, and it justifies a lighting assessment of the channel whatever the answer.

Whether the floodlighting affected the master's ability to detect the beacon was about as likely as not. It remains unresolved and is not carried into the findings.

**Water level below prediction reduced the depth over Boolarong Bank**

This was ruled out. The Kanooka Heads gauge recorded 0.91 m at 18:12, against a predicted 0.94 m. That left 1.41 m of water over the bed at the contact point, against a static draught of 1.35 m plus 0.18 m of squat. The vessel grounded because it was 54 m outside the dredged channel, so a lower water level was very unlikely.

**Day nine of twelve, on a split duty across a thirteen hour forty spread**

The master was on his ninth consecutive day, on a split duty of 9 hours 10 minutes across a spread of 13 hours 40 minutes. He had recorded 6 hours 15 minutes of sleep and described himself as tired. The duty was within every applicable limit.

The recorded evidence showed no sign of fatigue on the night:

- the centreline was held closely on entry;
- the response to the shallow alarm took 4 seconds, against a fleet figure of 6 to 9;
- the radio work was complete at the first attempt;
- the decision not to refloat was correct.

The decision fatigue would have had to degrade was stopping the vessel. Both masters said independently that no document prompts that decision. The outgoing master, who was not tired, described the same difficulty with the beacon.

It was unlikely that the duty pattern influenced this occurrence. It is likely to matter in future, because it recurs on every winter darkness transit. It is therefore reported as an other risk factor.

**Individual Actions**

## **Plotter fault passed on verbally at the change of master and not entered in the defect book**

The plotter froze twice during the morning shift, at 10:42 and 11:17, and recovered by itself each time. At the handover at about 13:05 the outgoing master told the relieving master about both freezes. The relieving master said the unit had done the same to him the month before. Neither master entered the fault in the defect book or reported it to the Operations Manager. Their accounts agree, and both volunteered the omission before they were asked.

Nothing the operator required was omitted. OPS-9 at 9.3 requires the outgoing master to brief the relieving master on "any defect". It does not require the brief to be written down. It does not require the relieving master to confirm that a defect mentioned in the brief has been entered. A non-statutory item falls under 9.2, due "at the end of the shift in which they are found". For a fault that clears itself, that provides no trigger. This analysis concerns the practice, not the two masters. The procedure behind the practice is argued under Risk Controls.

An entry would have put the fault in front of the people who buy firmware and withdraw vessels from service. The fault at 18:07:06 carried cause code 0x51. The manufacturer's firmware of 11/2024 addresses that code. The operator held no plotter entry for 2026 and no subscription to the manufacturer's bulletins, and the update had never been applied.

Two things limit the link. The bench test did not reproduce the fault, so the connection between the morning freezes and the evening freeze rests on the shared cause code. A single entry made at 13:05 would also have been reviewed on a weekly cycle, and would not by itself have taken TF18 out of service that evening.

Passing the fault on verbally, without an entry in the defect book, likely contributed to the plotter remaining in service with an unrectified fault that recurred during the transit.

## **Transit continued at eight knots after the plotter froze, rather than stopping in the channel**

The engine recorder shows no throttle movement from the freeze until 18:11:52. Speed over the ground fell from 8.4 kt to 7.6 kt by 18:11 with power unchanged. At 18:08:31, 85 seconds after the freeze, the bridge audio records the master turning to the search for the No. 4 beacon: "Right. Where's four." Nothing on the audio indicates that stopping was considered. The master's account agrees. He did not consider stopping. He went straight to the visual transit because he regarded it as the backup. He would have said beforehand that he would stop.

The conditions favoured stopping. There was no other traffic, the water was flat and the wind was offshore at 8 kt. The vessel's crash stop from 8 kt is 92 m. It crossed the western edge of the channel 114 seconds after the freeze and grounded 54 m outside it. A stop begun at any point in that window would have brought the vessel up short of the bank.

During the same period the heading was brought from 195 to 200 degrees and held. The vessel set west at an average of 12 m per minute with no means of fixing its position. The master was alone on the bridge. The deckhand had been sent to collect fares at 18:04:14, which was normal practice on the route and is not addressed by OPS-3.

The decision was shaped by conditions argued elsewhere. The passage plan named the visual transit as the secondary means of fixing and gave no hold point for loss of the plotter (Risk Controls). The beacon the master turned to was extremely likely not visible from the channel over the stretch he searched (Local Conditions). Reverting to the backup the plan named followed from those conditions. What removed the margin was continuing at 8 kt while searching.

Continuing the transit at eight knots after the plotter froze extremely likely contributed to the vessel leaving the dredged channel, and to the shoaling not being detected until 16 seconds before contact.

### **Engines stopped and no attempt made to refloat on a falling tide**

The master stopped both engines at 18:12:31, 27 seconds after contact, and they were not restarted. His pan pan at 18:13:04 stated "I am not attempting to refloat" as a decision already made. He addressed the passengers 36 seconds after the impact and again at 18:14:52. He had the No. 2 starboard void sounded, reported about 75 mm of ingress to Port Control at 18:17:05, and had the void sounded every ten minutes after that. All 38 passengers were issued lifejackets. The delay in the forward saloon is a separate matter, argued under Risk Controls. In interview the master gave four reasons: the falling tide, the unknown bottom, the number of people aboard, and that the vessel could not sink while it remained on the bank.

The survey confirmed the vessel was holed, with a 180 mm split in the starboard plating in way of the No. 2 void. Ingress reached about 260 mm by the time the vessel refloated at 00:20. The breach was in the compartment the vessel was resting on. Backing off into 3.2 m of water at night, with the ingress unassessed, would have put a flooding compartment into deep water with the passengers still aboard.

The decision to stop the engines and not attempt to refloat, made within about a minute of contact, very likely reduced the risk to the passengers and the vessel.

## **Occurrence Events**

### **Steering or propulsion failure caused the vessel to sheer off track**

A steering or propulsion failure was very unlikely. The engine and steering recorders showed normal values throughout the transit, the turn was a heading applied and then held rather than a sheer, and the steering system was found serviceable when it was removed and tested after the occurrence.

### **Vessel left the dredged channel and tracked onto Boolarong Bank**

It was virtually certain that the vessel left the dredged channel. The recorded AIS track, the surveyed grounding position and the furrows in the bed all place the vessel in the same position.

The track is precise about when the vessel left the centreline. The vessel was 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30 and held that line until 18:07:06. The plotter event log records the display freeze at 18:07:06, so the departure from the centreline began at the same moment as the freeze. From then the heading increased from 195 to 200 degrees and was held. The rate of turn peaked at 2.1 degrees per minute at 18:07:40 and was back to zero by 18:08:50.

The rate-of-turn profile is a heading change that was applied and then kept. It is not a drift. The wind was offshore at 8 kt, and there is no cross-set in the channel at that state of tide. Neither could produce a turn that starts, peaks and stops. On the held heading the vessel set west at an average of 12 m per minute. It reached the western edge of the 40 m channel at about 18:09:00 and was over Boolarong Bank from about 18:11:00. It made contact 74 m west of the centreline, 54 m outside the channel and 34 m west of the No. 4 beacon.

The bridge audio places the master's search for the No. 4 beacon inside this interval. He was looking for it at 18:08:31. At 18:10:02 he said "There's four. Keep her over." At that point the vessel was already 39 m west of the centreline and outside the dredged channel. The reasons the beacon could not be seen over that stretch are argued under Local Conditions. The possible effect of the foreshore floodlighting was examined but could not be resolved, and it is discussed in its own

subsection. Nothing from it is carried into this conclusion. The decision to continue at eight knots after the freeze is argued under Individual Actions.

The channel is dredged to 3.2 m. The grounding required the vessel to be outside it, and it followed from nothing else. This factor's contribution to the grounding was also virtually certain.

The vessel left the dredged channel on a heading that was deliberately applied and held from the moment the plotter froze, and it tracked west onto Boolarong Bank. This was virtually certain, and it was virtually certain to have contributed to the grounding.

### **Shoaling not detected until sixteen seconds before contact**

It was virtually certain that the shoaling was not detected until sixteen seconds before contact. The alarm log, the bridge audio and the recorded speed agree to the second.

The echo sounder's shallow alarm was set to 0.9 m under the keel. That value had been entered at commissioning in 2019 and never changed. The alarm sounded once on the voyage, at 18:11:48. Both throttles went to astern together 4 seconds later. Contact followed at 18:12:04, while astern thrust was still developing.

The alarm log gives the arithmetic: "At 7.1 kt over the ground the vessel covered 58 m in those 16 seconds." The vessel's trials figure for a crash stop from 8 kt is 92 m. The warning therefore gave less distance than the vessel needed to stop.

The equipment worked as set. The survey measured the bank gradient across the contact section at about 1 in 70. The alarm log records that on this gradient the alarm "could not have sounded appreciably earlier at that setting." The shortfall therefore lay in the combination of the setting and the speed, not in the echo sounder.

A setting that gave about forty seconds' warning would have allowed the vessel to stop. So would a transit speed at which sixteen seconds covered less than the stopping distance. In either case the vessel would have stopped short of the bank.

Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement. The engine recorder attributes this to drag from the shoaling water. The speed loss was not large enough to change the outcome.

The shoaling was not detected until sixteen seconds before contact, which was virtually certain. It was extremely likely to have contributed to the grounding, because a 0.9 m alarm setting on a 1 in 70 gradient at about eight knots gave less warning than the 92 m the vessel needed to stop.

## Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

## Findings

### Contributing safety factors

1. The winter timetable very likely moved the last service into darkness without the route risk assessment being reopened. This likely contributed to the grounding because RRA-07's daylight-only restriction, and its requirement for approval and a specific passage plan for any night transit, were never applied to service TF18. (Safety issue)
2. It was virtually certain that the operator had no night passage plan for the Weerona Channel and no hold point for the loss of the primary means of fixing. This absence very likely contributed, because nothing prompted the master to stop in the 114 seconds between the plotter freezing and the vessel crossing the channel's western edge. (Safety issue)
3. After the plotter froze, the transit continued at about 8 kt rather than stopping in the channel. This was virtually certain, and it very likely contributed by carrying the vessel out of the channel with no means of fixing its position, although a stop was available in 92 m throughout.
4. Shoaling was, with virtual certainty, not detected until 16 seconds before contact. This very likely contributed to the grounding: the shallow alarm, set at 0.9 m under the keel since 2019, worked as set but gave 58 m of warning against a crash-stop distance of 92 m.
5. The change-of-master defect brief was, with virtual certainty, verbal, with nothing checking that a defect mentioned in it was written down. This likely contributed, because three or four plotter freezes over a year never reached anyone ashore and the firmware issued in 11/2024 was never applied. The evidence also supports a second reading: an entry made that afternoon would have been reviewed at the end of the week, TF18 would have sailed either way, and on that reading the rule is a fleet-wide deficiency rather than a cause of this grounding. (Safety issue)
6. It was virtually certain that the plotter fault was passed on verbally at the change of master at about 13:05 and not entered in the defect book. The omission likely contributed by keeping an intermittent fault on the primary means of fixing from the people who could apply the firmware fix or withdraw the vessel. A single entry made that afternoon would not by itself have stopped TF18.
7. The chart plotter display very likely froze at 18:07:06 during the channel transit and did not recover. This is established by the unit's own event log, the bridge audio and the master's account, although the fault did not recur on a 46-hour bench test. The freeze very likely contributed by removing the primary means of position fixing named in the passage plan.
8. Under WAN-2, aids to navigation on the Weerona Channel carried no availability standard and no duty to tell users anything; this was virtually certain. It likely contributed, the least firmly held of the contributing findings. The notification duty in section 4.2 was the one mechanism that would have put the beacon's range change before the operator. However, an availability target would not have detected a light that was lit but screened, and nothing in the code prevented the Authority from notifying operators, trimming the mangroves or measuring the light. (Safety issue)
9. The No. 4 beacon was converted to solar in 11/2025, its nominal range was halved from 6 M to 3 M, and no operator was told. All of this was virtually certain. The conversion likely contributed, since the passage plan the master was working to still showed 6 M and the measured range was

2.6 nm. It is not rated higher because, over the stretch where the master was looking, the light was screened entirely. (Safety issue)

10. The No. 4 beacon was very likely not visible from the channel between 0.84 nm and 0.34 nm north of it, the half mile before the vessel left the channel. This was very likely a contributor: with the plotter frozen, the beacon was the only means of fixing available, and it was absent for the whole period in which the vessel set west.

11. With virtual certainty, the vessel left the dredged channel on a held heading change and tracked onto Boolarong Bank. This virtually certainly contributed, because the grounding position lay 54 m outside a 40 m wide channel dredged to 3.2 m.

12. Clearing of the beacon's sightline was virtually certainly deferred as unfunded at three works rounds, in 2023, 2024 and 2025. The deferral very likely contributed, because the Weerona Islet canopy grew to 5.8 m against a lantern height of 4.2 m and closed the northern sightline. (Safety issue)

### **Other findings**

13. The forward saloon lifejacket locker was, with virtual certainty, not reachable from 22 of the 34 seats it serves without passing behind the kiosk counter. The counter was fitted in a 2019 refit that did not trigger a re-survey, and the same counter is on the operator's two other 24 m vessels. The arrangement was not found to have influenced the grounding or its outcome, but it very likely increased risk to passengers in any emergency that required the saloon to equip itself. (Safety issue)

14. The master was, with virtual certainty, on day nine of twelve, on a split duty with a 13-hour 40-minute spread that ended with a darkness transit. The recorded helm work, alarm response and post-grounding radio work gave no sign that fatigue influenced this occurrence. The roster likely increased risk as a standing exposure on every winter darkness transit of the route.

15. A water level below prediction was very unlikely to have reduced the depth over Boolarong Bank. The gauge read 0.91 m at 18:12 against a prediction of 0.94 m, and this hypothesis was ruled out.

16. A steering or propulsion failure was very unlikely. Recorded engine and steering parameters were normal throughout, and every steering component tested serviceable. This hypothesis was ruled out.

17. After contact the master stopped both engines and made no attempt to refloat on a falling tide with a breached void. This very likely reduced risk to the 38 passengers on board.

The investigation examined whether new foreshore floodlighting behind the No. 4 beacon reduced the master's ability to pick it up, and could not resolve it. Fourteen unshielded floodlights, commissioned on 22/05/2026, lie within 4° of the beacon's bearing from 0.5 nm north. Both masters independently described the beacon as washed out against them. However, that alignment falls inside the arc where the canopy screened the light entirely. The master acquired the beacon at 0.34 nm with the floodlights unchanged. No measurement exists from the night itself. Whether the floodlights affected detection could not be established while the mangroves screen the beacon. Settling the question would require a detection assessment of the light against that background with the sightline clear.

### **Probable cause**

The probable cause was that the chart plotter display very likely froze during a night transit for which the operator had no night passage plan and no hold point. The vessel then continued at about 8 kt while the No. 4 beacon, the only remaining means of fixing, was very likely screened

from the channel by mangrove canopy. As a result, the vessel left the dredged channel and tracked onto Boolarong Bank, where the shallow alarm gave 16 seconds' warning against a 92 m stopping distance.

## Contributing factors

- The winter timetable change very likely put TF18 into darkness without the route risk assessment being reopened, and this likely contributed.
- The change-of-master defect brief was, with virtual certainty, verbal and unchecked. This likely contributed, although it can equally be read as a fleet-wide deficiency rather than a cause of this grounding.
- It was virtually certain that the plotter fault was not entered in the defect book at the change of master, and this likely contributed.
- WAN-2 placed no availability, recording or notification duty on the Weerona Channel's aids, which was virtually certain. This likely contributed, though it was not firmly established.
- The No. 4 beacon's solar conversion halved its nominal range without any operator being told; this was virtually certain and likely contributed.
- Clearing of the No. 4 beacon's sightline was deferred as unfunded for three works rounds, which was virtually certain and very likely contributed.

## Recommendations

### **Aids to navigation on this waterway carry no availability standard and no duty to tell users anything**

The Weerona Channel is not a declared shipping channel. Section 4.2 of Waterways Aids to Navigation Code WAN-2 sets three requirements that apply only to aids on declared shipping channels: an availability target of 99.0 per cent, an obligation to record availability, and an obligation to publish a notice to mariners when an aid's character, position or range changes. For this channel, section 4.3 requires only that the responsible authority can show an inspection interval it has set itself.

The services agreement between Kanooka Shire Council and Kanooka Heads Port Authority sets that interval. It provides for:

- a quarterly daylight inspection
- a monthly observation from the town wharf
- vegetation work no further than 50 m from the aid.

The agreement imposes no duty to tell any waterway user about a defect, a deferral or a change. The Port Authority was not in breach of either instrument.

This scope likely contributed to two outcomes. The halving of the No. 4 beacon's nominal range after its solar conversion reached no operator. The clearing of the beacon's sightline was deferred through three works rounds. The notice-to-mariners duty was the one mechanism that would have put the range change in front of the operator, whose passage plan still carried the earlier figure on the night.

The investigation did not regard this influence as settled. Nothing in the code prevented the Port Authority from notifying operators, trimming the mangroves or measuring the light. An availability target would also not have detected the problem, because the lantern was lit throughout and the problem was that it could not be seen.

No safety action has been recorded against this issue, and no recommendation is made.

### **Beacon converted to solar, nominal range halved, and no operator was told**

In November 2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal 3 nautical miles, against 6 for the lantern it replaced. The character, Fl G 2.5s, did not change. The investigator measured a luminous range of 2.6 nm from the port workboat on 18/07/2026.

A notification was lodged with the hydrographic service on 04/02/2026. The chart correction was still pending on the night of the grounding. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix, issue 2 and current, still recorded the beacon at 6 M. That this condition existed was virtually certain.

The master was navigating to a written plan that described a light more than twice as strong as the one in front of him. A weaker light is harder to acquire, and the master spent 91 seconds failing to acquire it. The reduced range likely contributed to the beacon not being visible from the channel over the half mile before the vessel left it. It was not rated higher because the light is screened entirely over the same stretch where the master was looking, and a brighter light would not have overcome that screening.

The condition arose from the absence of any availability standard for aids to navigation on this waterway, and of any duty to inform users of changes to them.

No safety action has been recorded against this issue.

### **Clearing of the beacon's sightline deferred unfunded for three works rounds**

Mangrove regrowth on Weerona Islet screened the No. 4 beacon from the northern approach, and the Port Authority deferred clearing it at three successive works rounds. It is virtually certain this occurred.

Work order WO-118, raised in 09/2023, recorded the screening. It recommended trimming the canopy to 2 m on a three-yearly cycle, at an estimated \$18,500. The work was deferred at the 2023, 2024 and 2025 works rounds. Each entry read 'defer, unfunded, refer next round'. WO-176, raised in 03/2026, recorded the sightline as 'now fully closed on the northern approach'. The Authority's file note against WO-176 recorded that the islet is 240 m from the beacon. That places it outside the 50 m scope of schedule 2.4, so the work competed for the general waterways vote rather than the aids to navigation vote. The canopy measured 2.1 m in the 2019 aerial survey and 5.8 m on 18/07/2026. The lantern height is 4.2 m.

The issue matters because the screening removed the beacon from view rather than dimming it. On 18/07/2026 the screened arc ran from 0.84 nm to 0.34 nm north of the beacon, and the vessel left the channel inside that arc. The deferred clearing very likely contributed to the beacon not being visible over that stretch. The allowance against a higher grading reflects the possibility that the master might not have picked up the light in that minute and a half even had it been visible.

No safety action has been recorded against this issue, and no recommendation is made.

### **Winter timetable moved the last service into darkness without reopening the route risk assessment**

The winter timetable moved service TF18 from 16:15 to 17:35 from 01/06/2026, which placed the last service of the day in darkness. RRA-07 issue 2, the route risk assessment for the Weerona Channel, restricted the channel to daylight transits. It required the General Manager Operations' approval and a specific passage plan for any night transit. On 22/04/2026 the Operations Manager approved change record CHG-2026-014. The change record assessed the change as a timetable change only, identified no new hazard, and required no review of the route risk assessment. RRA-07 was not reopened. No night passage plan was written, and the General Manager Operations' approval was neither sought nor given. It was extremely likely that the change was made in this way.

TF18 then ran 31 times in darkness before the occurrence. Each time, a 96-passenger vessel transited a 40 m channel, contrary to a restriction that was still in force. The change process asked whether the change was new, not whether the existing assessment still held. It was likely that this contributed to the occurrence. A review of RRA-07 against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons. That reach was already identified as a hazard, and the review would have required a night passage plan before the first service ran. Such a review might still have produced a plan no more informative than the daylight one.

No safety action has been recorded against this issue.

### **No night passage plan for the channel, and no hold point if the primary means of fixing is lost**

The operator had no night passage plan for the Weerona Channel. Its passage plan named no hold point or abort criterion for a transit in which the primary means of position fixing was lost. This was virtually certain.

Appendix A to OPS-3 names the chart plotter as the primary means of fixing. The secondary means is the visual transit of the No. 4 beacon with the town wharf leading lights. The plan was written for the daylight transit and states that night transits are not scheduled. It says nothing about losing the plotter. OPS-3 3.6 requires a reduction of speed and consideration of anchoring only in reduced visibility, which did not apply on the night.

When the plotter froze, the leading lights had not yet resolved as two lights. The beacon was therefore the whole of the secondary means. The master had 114 seconds between the freeze and crossing the channel edge, on flat water with no other traffic. A named hold point would have been available throughout that time. It was very likely that the absence of a hold point contributed to the transit continuing at eight knots rather than stopping in the channel. Both masters interviewed said independently that the right response was to stop. Both also said that nothing in the operator's procedures tells a master to do so.

No safety action has been recorded against this issue, and this report makes no recommendation in respect of it.

### **Change-of-master defect brief is verbal, with nothing checking that what is said gets written down**

The operator's defect-reporting rules in Operations Manual OPS-9 required the outgoing master to brief the relieving master on the state of the vessel and any defect. They did not require that brief to be written down, and nothing required the relieving master to confirm that a defect mentioned in it had been booked. It was virtually certain that this arrangement existed. The manual text sets it out, and both masters independently described the same practice: the book was for what was broken at handover, and what had come good was spoken about.

Under these rules the chart plotter was not a statutory item. It was due in the defect book only at the end of the shift in which a fault was found, even though it was the primary means of position fixing named in the passage plan. For an intermittent fault, that requirement never took effect.

The arrangement likely contributed to the grounding. Three or four plotter freezes over the preceding year never reached anyone ashore who could act on them. As a result, the firmware fix issued in 11/2024 was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance.

The investigation also considered a contrary reading. An entry made at 13:05 on the day would have been reviewed by the Operations Manager at the end of that week, and the vessel would have sailed TF18 either way. On that reading the rule is a weakness across the fleet rather than a cause of this grounding. On either reading, the rule explains why the fault went unrecorded. It does not excuse the omission.

No safety action has been recorded against this issue.

### **Forward saloon lifejacket locker not reachable from most of the saloon it serves**

On MV Tarrakin Star, the forward saloon lifejacket locker could not be reached from most of the saloon it served. A survey on 16/07/2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the kiosk counter. With people standing at the counter, a 0.6 m inboard gap was the only route. The muster placard directed passengers to a

locker they could not see from their seats. When the vessel was built in 2011, the approved arrangement had an open bench in that position, and the locker face was clear. The counter was fitted during a 2019 interior refit under a minor modification notice. That notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

It is unlikely that the locker's position had any bearing on the grounding or its outcome. All 38 passengers were in lifejackets within about eight minutes and ashore within 74. The vessel was aground, upright, stable and not sinking.

The arrangement very likely increased risk. The following circumstances could each make the locker's position decisive:

- a fire
- a risk of capsize
- a night evacuation into the water
- a deckhand injured in the impact

In any of these, whether the saloon's passengers can equip themselves or must wait for crew help depends on the locker. The underlying issue is the modification route. An interior refit changed a safety equipment arrangement, and nothing in that route checked whether it had.

No safety action has been recorded against this issue.

## Distribution

| Title                                             | Name         |
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